Statistical Rigor & Diagnostics Focus

Logistic Regression for Heart Disease Prediction

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Abstract

This presentation provides a comprehensive guide to Statistical Rigor and Diagnostics in Logistic Regression, utilizing a clinical Heart Disease dataset.

- **Foundations:** Advanced data preparation, missingness mechanisms, and proper feature encoding.
- **Modeling:** Building and interpreting Logistic Regression models using Maximum Likelihood Estimation (MLE).
- **Diagnostics:** Rigorous testing of assumptions (Linearity of Logits, Multicollinearity via VIF, Influential Observations via Cook's Distance, and Interaction Effects).
- **Evaluation:** ROC-AUC, Precision-Recall curves, and optimal threshold selection using Youden's J statistic.

We emphasize the translation of statistical outputs into actionable clinical implications, highlighting the critical distinction between Odds Ratios and Risk Ratios.

Session Objectives

- Understand and handle missingness mechanisms (MCAR, MAR, MNAR) in clinical data.
- Apply proper feature encoding and assess multicollinearity using Variance Inflation Factor (VIF).
- Formulate, estimate, and interpret Logistic Regression models using `statsmodels`.
- Calculate and interpret Adjusted Odds Ratios with 95% Confidence Intervals.
- Diagnose model assumptions: Linearity of Logits, Influential Observations, and Interaction Effects.
- Evaluate predictive performance using ROC-AUC and Precision-Recall curves.
- Determine optimal clinical policy thresholds using Youden's J statistic.

Part 1: Advanced Data Prep

"Laying the groundwork for rigorous inference."

Target Audience, Dataset & Dependencies

- **Target Audience:** Analysts transitioning to rigorous statistical modeling.
- **Dataset:** `Heart.csv` (Clinical cardiovascular data).
- **Dependencies:**
  - Data Manipulation: `pandas`, `numpy`
  - Statistical Modeling: `statsmodels`
  - Machine Learning: `scikit-learn`
  - Visualization: `matplotlib`, `seaborn`

Missingness Mechanisms

Understanding Missing Data

- **MCAR (Missing Completely at Random):** Missingness is unrelated to any observed or unobserved data.
- **MAR (Missing at Random):** Missingness is related to observed data (e.g., older patients might skip the `Ca` test).
- **MNAR (Missing Not at Random):** Missingness is related to the unobserved value itself (e.g., patients with severe disease avoid the `Thal` scan).

Clinical Reality

In clinical data, MAR or MNAR is common. We must impute cautiously to avoid introducing bias.

Missingness Audit & Imputation Strategy

- **Audit:** `Ca` (4 missing), `Thal` (2 missing).

Variable	Type	Imputation Strategy
`Ca`	Ordinal (0–3)	Median imputation, robust to outliers.
`Thal`	Nominal	Create an explicit `unknown` category to preserve information about missingness and reduce potential bias.

Proper Feature Encoding

Nominal Variables
Use Dummy Encoding.
`pd.get_dummies(..., drop_first=True)`

Why drop_first? To avoid the dummy variable trap (perfect multicollinearity) in `statsmodels`.

Ordinal Variables
Ensure they are treated as numeric.

Examples:

- `Slope` is already 1, 2, 3.
- `Ca` is now numeric after imputation.

`statsmodels` will handle them as continuous or explicitly categorical depending on the formula.

Checking for Multicollinearity (VIF)

- We check core continuous predictors: `Age`, `RestBP`, `Chol`, `MaxHR`, `Oldpeak`.
- High VIF (> 5 or > 10) indicates problematic multicollinearity.

Feature	**VIF**
Age	1.347
RestBP	1.132
Chol	1.059
MaxHR	1.323
Oldpeak	1.175

Interpretation

Max VIF is 1.35 (Age). Since all values are well below 5, **no severe multicollinearity** is detected. No need to drop or combine variables.

Part 2: Building & Interpreting the Logistic Model

"From formula to meaningful clinical coefficients."

Defining the Statistical Formula

The Model Specification

```
HD ~ Age + Sex + C(ChestPain, Treatment('typical')) + RestBP + Chol + Fbs + C(RestECG) + MaxHR + ExAng
+ Oldpeak + C(Slope) + Ca + C(Thal, Treatment('normal'))
```

- `C()` denotes categorical variables.
- `Treatment('...')` explicitly sets the reference category for interpretation.
- `statsmodels` will automatically handle the dummy encoding based on this formula.

Fitting the Logistic Regression Model

- **Method:** Maximum Likelihood Estimation (MLE).
- **Implementation:** `smf.logit(formula, data=df).fit(disp=0)`
- `disp=0` suppresses verbose convergence output for cleaner reporting.
- **Note:** Missing values were dropped (`df.dropna()`) prior to this specific model fit for demonstration, yielding 297 observations.

Model Summary Output

Metric	Value	Metric	Value
Dep. Variable	HD	No. Observations	297
Model	Logit	Df Residuals	278
Method	MLE	Pseudo R-squ.	0.5325
Log-Likelihood	-95.821	LLR p-value	2.124×10^{-36}

Interpretation

The LLR p-value (< 0.001) indicates the model fits the data significantly better than an intercept-only (null) model. A Pseudo R-squared of 0.5325 indicates a strong fit for logistic regression.

Interpreting Coefficients and Odds Ratios

- **Coefficient (β):** Represents the change in the *log-odds* of the outcome for a one-unit change in the predictor.
- **Odds Ratio (OR):** Calculated as $\exp(\beta)$.
  - **OR $>$ 1:** Increased odds of the outcome.
  - **OR $<$ 1:** Decreased odds of the outcome.
  - **OR $=$ 1:** No effect.
- We evaluate statistical significance using the 95% Confidence Interval (CI) and p-value.

Adjusted Odds Ratios (Top Predictors)

Predictor	Coef	OR	95% CI	P-value
ChestPain (asymptomatic)	2.086	8.057	[2.18, 29.75]	0.002
Sex	1.546	4.693	[1.66, 13.26]	0.004
Thal (reversable)	1.393	4.026	[1.75, 9.26]	0.001
Ca	1.312	3.712	[2.15, 6.42]	0.000

- **Interpretation:** Patients with asymptomatic chest pain have 8.06 times the *odds* of heart disease compared to those with typical chest pain, holding all else constant.

CRUCIAL CONCEPT: Odds Ratios (OR) vs. Risk Ratios (RR)

Definitions

- **Odds Ratio:** Odds of outcome given exposure / Odds of outcome without exposure. (Native to Logistic Regression).
- **Risk Ratio (Relative Risk):** Probability of outcome in exposed / Probability in unexposed.

The Prevalence Trap

If the outcome is rare ($< 10\%$), OR $\approx$ RR.

However, Heart Disease in this dataset is $\sim$**50% prevalent**. Therefore, the OR will **exaggerate** the effect size compared to the RR.

Best Practice

For healthcare reporting, do not say "8 times more likely." Instead, convert predicted probabilities to marginal risk differences or explicitly state that OR is not RR.

Part 3: Statistical Diagnostics

"Ensuring the validity of our statistical inferences."

Testing Linearity of Logits

- **Assumption:** The relationship between continuous predictors and the log-odds of the outcome is linear.
- **Method:** Box-Tidwell concept.
- **Practical Diagnostic:**
  1. Bin a continuous variable (e.g., `Age` into quintiles).
  2. Calculate the mean `Age` and mean `HD` per bin.
  3. Calculate the logit: $\ln(\frac{p}{1-p})$ for each bin.
  4. Plot Mean Age vs. Logit. A straight line indicates the assumption holds.

Linearity of Logit Results

Age Bin	Mean Age	Mean HD (p)	Logit
0	41.25	0.246	-1.121
1	49.89	0.344	-0.644
2	56.19	0.557	0.230
3	60.36	0.711	0.901
4	66.52	0.500	-0.000

Visualization

A regression plot of these points (saved as `linearity_of_logit_age.png`) shows a generally linear trend, confirming the assumption holds reasonably well for Age.

Identifying Influential Observations

- Some data points may exert disproportionate leverage on the regression coefficients.
- **Metrics:**
  - **Cook's Distance:** Measures the effect of deleting a given observation.
  - **Leverage (Hat matrix diagonal):** Measures how far an observation's predictor values are from the mean.
- These diagnostics are extracted via `model.get_influence()` and added back to the dataframe for inspection.

Influential Observations Results

- **Rule of Thumb:** Cook's Distance $> 4/n$ is potentially influential.
- **Threshold:** $4/297 \approx 0.0135$
- **Finding:** 29 potentially influential observations identified.

Obs	Age	Sex	ChestPain	HD	Cook's D
12	56	1	nonanginal	1	0.0183
16	48	1	nontypical	1	0.0404
32	64	1	nonanginal	1	0.0314

Best Practice

Do not blindly delete outliers. Investigate them first. If they are data entry errors, remove them. If genuine, consider robust regression techniques.

Testing for Confounding and Interaction Effects

- **Question:** Does the effect of `ExAng` (Exercise Induced Angina) on heart disease differ by `Sex`?
- **Method:** Add an interaction term (`Sex:ExAng`) to the model.
- **Formula:** `HD ~ Age + Sex + ExAng + Sex:ExAng`
- If the p-value for the interaction term is < 0.05, the effect of `ExAng` significantly differs by `Sex` (effect modification).

Interaction Effect Check Results

Predictor	Coef	Std. Err	z	P>|z|	95% CI
Intercept	-5.661	1.035	-5.467	0.000	[-7.69, -3.63]
Age	0.068	0.016	4.128	0.000	[0.036, 0.100]
Sex	1.691	0.396	4.274	0.000	[0.916, 2.467]
ExAng	2.464	0.572	4.307	0.000	[1.342, 3.585]
Sex:ExAng	**-0.792**	**0.670**	**-1.181**	**0.238**	**[-2.105, 0.522]**

Interpretation

The p-value for `Sex:ExAng` is 0.238 (> 0.05). There is **insufficient evidence** that the effect of Exercise Induced Angina on heart disease differs significantly according to Sex.

Part 4: Evaluation & Thresholding

"Translating probabilities into clinical policy."

Train/Test Split Strategy

- **Purpose:** Avoid overly optimistic in-sample metrics and assess generalizability.
- **Implementation:**
  - `test_size=0.2` (80% train, 20% test)
  - `random_state=42` (Reproducibility)
  - `stratify=y` (Maintains the proportion of Heart Disease cases in both splits)

Note on Demonstration

For the simplicity of visualizing curve shapes in this script, full dataset probabilities are used. In strict production, predictions must be mapped exclusively through `X_test`.

ROC-AUC Curve

Receiver Operating Characteristic
Plots True Positive Rate (Sensitivity) vs. False Positive Rate (1 - Specificity) across all thresholds. **Result:**

ROC-AUC = 0.935 Interpretation:

Excellent ability to distinguish between patients with and without heart disease.

Visual Output

The curve (saved as `roc_curve.png`) bows sharply toward the top-left corner, far above the diagonal line of no-discrimination (AUC $= 0.5$).

Precision-Recall Curve

Why PR Curve?
Crucial for imbalanced or high-stakes clinical data where False Positives and False Negatives carry heavy, asymmetric costs.
Result:

PR-AUC = 0.929 Interpretation:

Demonstrates strong predictive performance in balancing Precision (Positive Predictive Value) and Recall (Sensitivity).

Clinical Relevance

In heart disease screening, a high Precision-Recall AUC ensures that when the model flags a patient as high-risk, it is highly likely to be a true positive, minimizing unnecessary, invasive follow-up procedures.

Threshold Selection for Clinical Policy

- **Default Threshold:** 0.5 (may not be optimal for clinical trade-offs).
- **Optimization Method:** Youden's J statistic.
- **Formula:** $J = \text{Sensitivity} + \text{Specificity} - 1$
- **Result:** Optimal Classification Threshold = **0.481**

Clinical Impact

At this threshold (0.481), we optimally balance False Positives (unnecessary angiograms) and False Negatives (missed, life-threatening diagnoses).

Part 5: Deliverable

"The Statistical Diagnostics Report."

Statistical Diagnostics Report: Prep & VIF

1. Data Preparation

- **Missingness:** `Ca` imputed with median (0.0); `Thal` imputed as `unknown`.
- **Encoding:** Dummy variables applied to nominal features; ordinal features retained.

2. Multicollinearity (VIF)

- **Max VIF observed:** 1.35 (`Age`).
- **Recommendation:** No severe multicollinearity detected. Model coefficients are stable.

Statistical Diagnostics Report: Model Fit & OR

3. Model Fit & Odds Ratios

- **Pseudo R-squared (McFadden):** 0.533 (Indicates substantial improvement over the null model).
- **Top 3 Predictors by Odds Ratio:**
  1. **ChestPain (asymptomatic):** $\mathrm{OR} = 8.06$, $p = 0.002$
  2. **Sex:** $\mathrm{OR} = 4.69$, $p = 0.004$
  3. **Thal (reversable):** $\mathrm{OR} = 4.03$, $p = 0.001$

Statistical Diagnostics Report: Diagnostics & Metrics

4. Diagnostics

- **Linearity of Logit:** Visually inspected and holds.
- **Influential Observations:** 29 records exceed Cook's D threshold of 0.0135 (warrants review).
- **Interaction Check:** `ExAng * Sex` interaction p-value = 0.2375 (not significant).

5. Evaluation Metrics

- **ROC-AUC:** 0.935
- **Precision-Recall AUC:** 0.929
- **Recommended Policy Threshold:** 0.481

Clinical Interpretation

NOTE ON CLINICAL REPORTING

Odds Ratios (OR) are reported above. Because the prevalence of Heart Disease in this cohort is high ($\sim$50%), the OR will **overstate** the Risk Ratio (RR).

Actionable Directive: Do not interpret OR as "X times more likely." Instead, use marginal predicted probabilities for patient-facing risk communication. These measures provide a more intuitive representation of an individual's estimated probability of having heart disease.

Overall Conclusion

- The diagnostic assessment indicates the logistic regression model has **good statistical properties** and **excellent predictive performance**.
- There was no evidence of problematic multicollinearity.
- Asymptomatic chest pain, Sex, and reversible Thal were among the strongest, statistically significant predictors.
- The non-significant `ExAng` $\times$ `Sex` interaction suggests no effect modification by Sex.
- **Next Step:** The 29 potentially influential observations should be investigated before finalizing the model for production deployment.

Q&A and Thank You!

Questions?

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